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Influence of Socio-Economic Factors on Student Performance

Introduction

Understanding the factors that influence student performance is crucial for educators, policymakers, and researchers. Socio-economic factors, including parental education, lunch type, and test preparation courses, play a significant role in shaping academic outcomes. To explore these relationships, we utilize a dataset containing detailed information on students' backgrounds and their performance in various subjects. By creating a network visualization using Gephi, we can uncover and illustrate the intricate web of interactions between these variables.

Data Overview

The dataset used in this study includes information on 1,000 students, encompassing the following variables:

Gender: Categorized as male or female.

Race/Ethnicity: Grouped into categories (e.g., group A, B, C, D, E).

Parental Level of Education: Ranges from some high school to master's degree.

Lunch Type: Standard or free/reduced, indicating socio-economic status.

Test Preparation Course: Completed or none.

Performance Scores: Scores in math, reading, and writing.

The primary objective is to analyze how these socio-economic factors correlate with and influence student performance in these subjects.

Network Visualization

Using Gephi, we created a network visualization to represent the relationships between the variables. In this visualization:

Nodes: Each node represents a different variable from the dataset.

Edges: Edges connect nodes, indicating relationships between variables. Thicker edges represent stronger correlations.

Key Insights from the Visualization

Central Role of Gender:

The nodes for "male" and "female" are centrally located, highlighting the significant role of gender in the dataset. These nodes are connected to numerous other variables, suggesting that gender differences influence a wide array of factors related to student performance. For instance, the performance in subjects like math and reading may vary between male and female students due to various underlying socio-cultural factors.

Impact of Parental Education:

Nodes representing different levels of parental education (some high school, high school, some college, associate's degree, bachelor's degree, and master's degree) are prominently connected to the performance score nodes. This indicates that higher levels of parental education are generally associated with better student performance. The influence of parental education could be due to several reasons, including the availability of resources, academic support at home, and parental involvement in their children's education.

Socio-Economic Status and Lunch Type:

The nodes for "standard" and "free/reduced" lunch types are significantly connected to performance scores. The lunch type serves as an indicator of socio-economic status, with free/reduced lunch often provided to students from lower-income families. The connections suggest that students receiving free/reduced lunch tend to have lower performance scores, highlighting the impact of socio-economic challenges on academic achievement.

Importance of Test Preparation:

The nodes "completed" and "none" for the test preparation course show clear links to performance scores. Students who completed test preparation courses generally perform better, indicating the effectiveness of these courses in enhancing academic outcomes. This underscores the importance of providing access to test preparation resources, especially for students from disadvantaged backgrounds.

Performance Scores:

Math, reading, and writing scores are linked to multiple socio-economic factors, demonstrating that student performance is influenced by a complex interplay of variables. For example, a student's math score might be influenced not just by their own study habits but also by their parents' education level, the type of lunch they receive, and whether they took a test preparation course.

Note: Filter 1 which is filtered by Giant Component in the topology section.

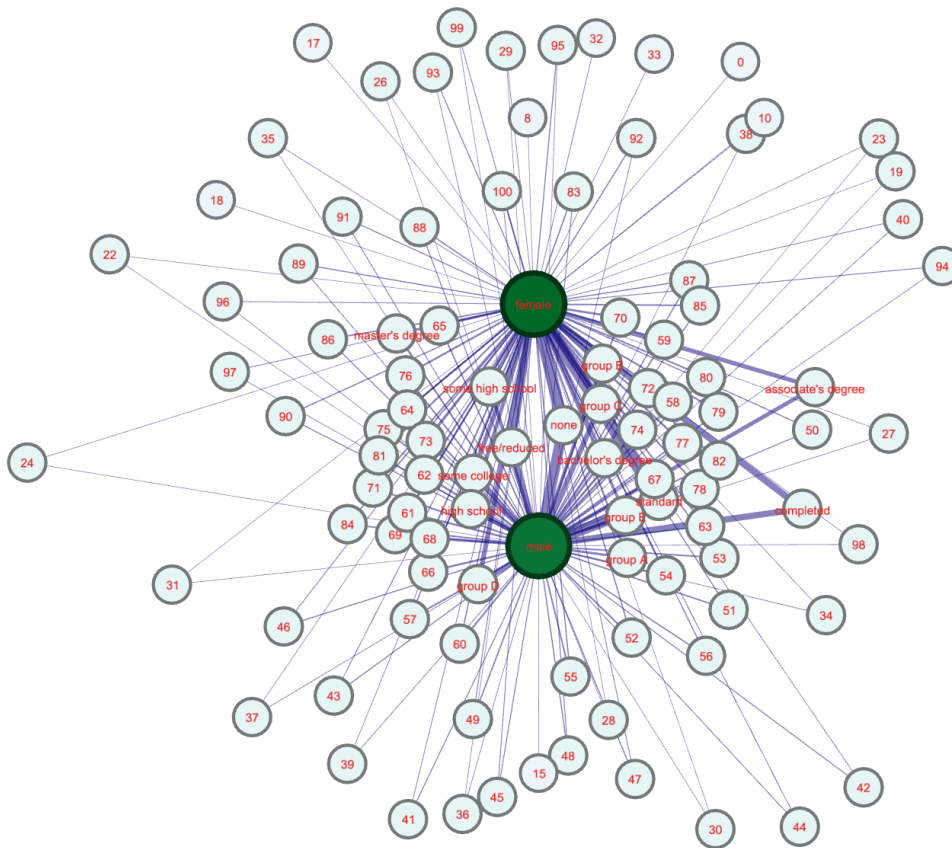


Figure 1: This network visualization illustrates the relationships between various socio-economic factors and student performance. Nodes represent different variables, such as gender, parental education, lunch type, and performance scores, while edges indicate the strength of correlations between them. Thicker edges signify stronger relationships. Central nodes, like those representing gender and parental education, highlight their significant influence on academic outcomes.

Note: Filter 3 which is filtered by degree range from 1 to 78 in the topology section.

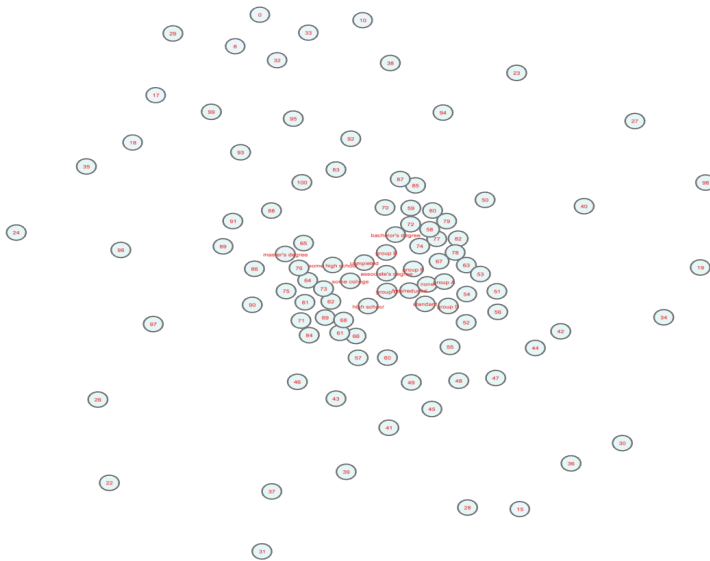


Figure 3: This network visualization showcases the socio-economic factors influencing student performance, filtered by degree range from 1 to 78 in the topology section. Each node represents a variable such as gender, parental education, lunch type, and performance scores, while edges indicate the strength of relationships between these variables. The filtering process focuses on nodes with degrees within the specified range, highlighting interactions with varying levels of connectivity and revealing underlying patterns within the dataset.

Analysis and Interpretation

The visualization reveals how interconnected socio-economic factors affect student performance. Larger nodes and thicker edges in the graph help identify the most influential variables and the strength of their relationships. For instance, the strong connection between parental education and student performance suggests that efforts to enhance parental engagement and educational attainment could positively impact student outcomes.

By examining these relationships, educators and policymakers can develop targeted interventions to support students. For example, providing additional resources and support to students from lower socio-economic backgrounds can help mitigate the disadvantages they face. Similarly, expanding access to test preparation courses can improve performance across the board, particularly for students who might not otherwise have the means to afford such resources.

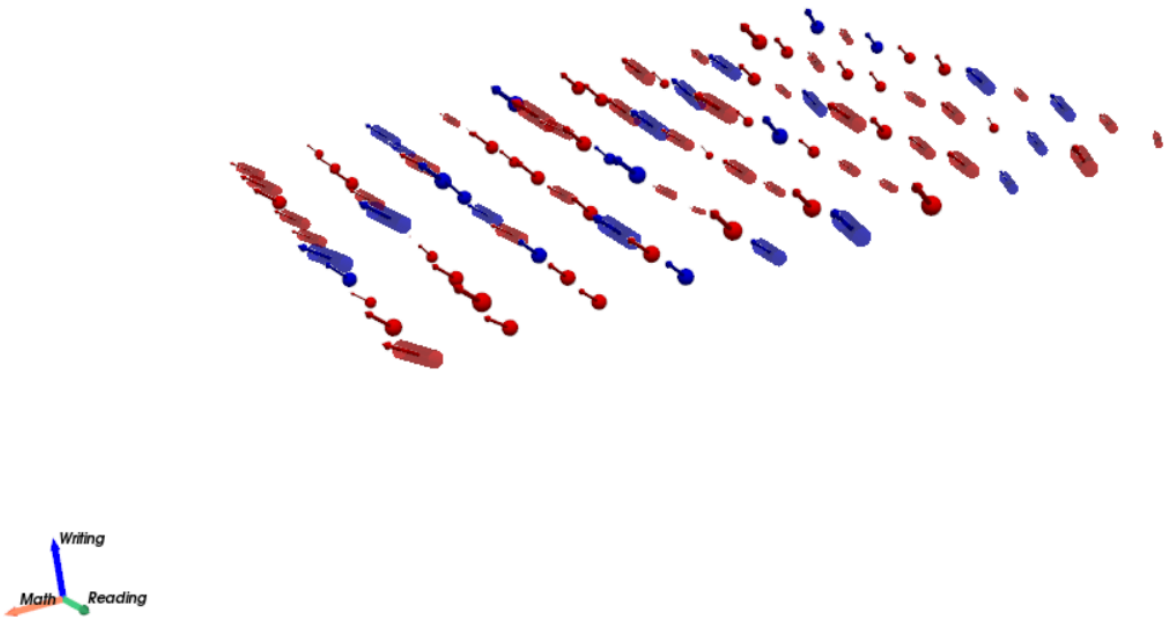


Figure 4: This glyph visualization illustrates the relationships between various socio-economic factors and student performance. Each aspect of the shapes is representative of a different value. Color represents prep course completion status (blue is completed, red is not), size represents the average of a student's test scores, and the base shape represents gender (spheres for females and cylinders for males). The shape direction is representative of the total weight of the student's test scores (x-axis is weighted for the math scores, y-axis for reading, and z-axis for writing).

Analysis and Interpretation

This visualization shows connections between different kinds of students and how different factors affect test scores. It's shown that all students generally have a similar direction as far as the overall weight of their exams. This is likely due to a large sample size and the data being averaged that causes the directions to be so similar. We can also see some large variation in test score averages in the large difference in sizes. Unsurprisingly, the students that did not complete a test preparation course had some of the lowest average scores. It's also shown here that female students had a tendency to perform better on average compared to males in this data set. With this figure there are many different kinds of connections we can draw based on the data set. Above are just a few of the more obvious connections that we are able to see at a first glance.

This visualization can help us understand the factors that contribute to a given student's successes or failures when it comes to standardized testing.

Uncertainty Visualization

In the realm of educational research, accurately understanding the degree of certainty surrounding the relationships between socio-economic factors and student performance is of paramount importance. Uncertainty visualizations serve as a vital tool by illustrating not only the connections within the data but also the confidence levels associated with these findings. By depicting uncertainty, these visualizations help researchers and policymakers identify which relationships are backed by robust data and which may require further investigation or additional data collection.

Designing Uncertainty Visualizations

Effective uncertainty visualizations are designed to make complex statistical concepts accessible and understandable. This involves encoding uncertainty in a clear and intuitive manner through visual modifications to traditional graphical representations. For instance, varying the thickness and color intensity of lines in a network graph can reflect the degree of uncertainty in the relationships. Additionally, the use of gradation in colors or line styles helps convey not just the presence of uncertainty but also its magnitude. Providing context within the visualization is crucial, as it ensures that viewers understand the nature and implications of the uncertainty depicted. This means including descriptive legends and explanatory notes that detail the sources of uncertainty and the statistical methods used to estimate it. Moreover, the visualization should allow for easy comparison between different data points or variables, highlighting their relative certainties and uncertainties, which can guide further research priorities or policy decisions. For example, while parental education shows a strong and consistent correlation with student performance, the impact of lunch type is less certain and shows greater variability across different schools or regions.

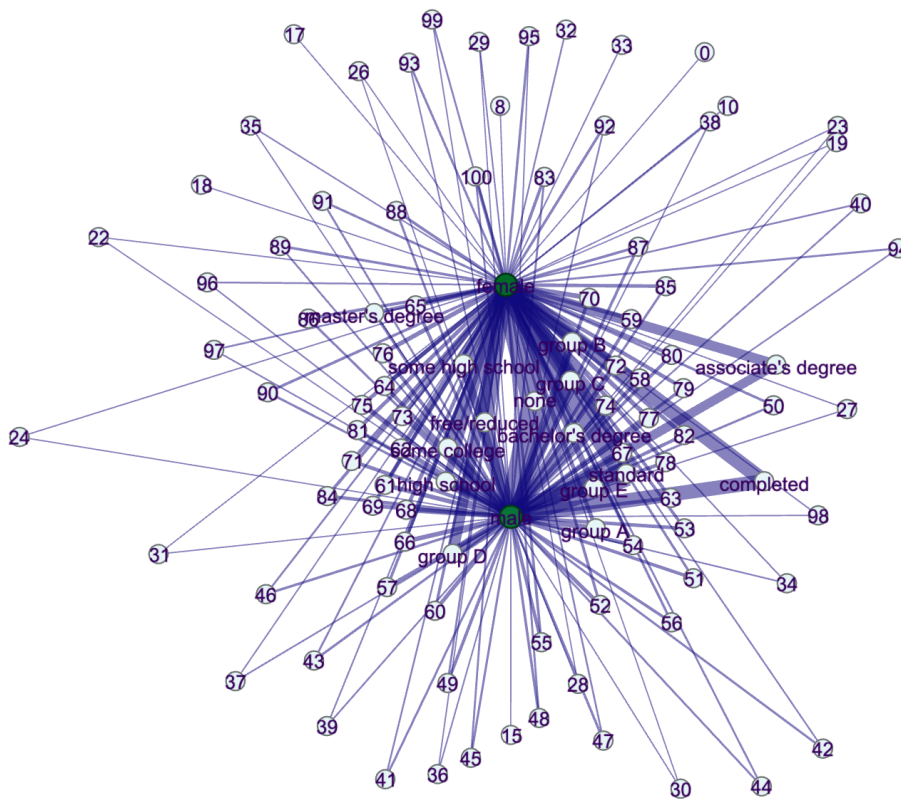


Figure 5: This graph illustrates the relationships between various socio-economic factors and their influence on student performance, with a focus on visualizing uncertainty. In this graph, nodes represent factors such as parental education levels, economic status, and access to educational resources, while the edges signify the relationships between these factors. The thickness and color of these edges vary, representing the degree of uncertainty in each relationship. Thicker, more vividly colored edges suggest a higher degree of confidence in the relationship, whereas thinner, fainter lines indicate less certainty.

This visualization helps highlight areas within the network where the data might be less reliable or where variability is greater. For instance, relationships involving less commonly studied variables or those with conflicting findings in the literature are marked with lighter, less distinct connections. This visual approach not only draws attention to the most and least certain aspects of the data but also serves as a call to action for focusing future research efforts on those areas with higher degrees of uncertainty. By doing so, educational researchers and policymakers can better allocate resources and design interventions that are based on the most reliable and comprehensive data available.

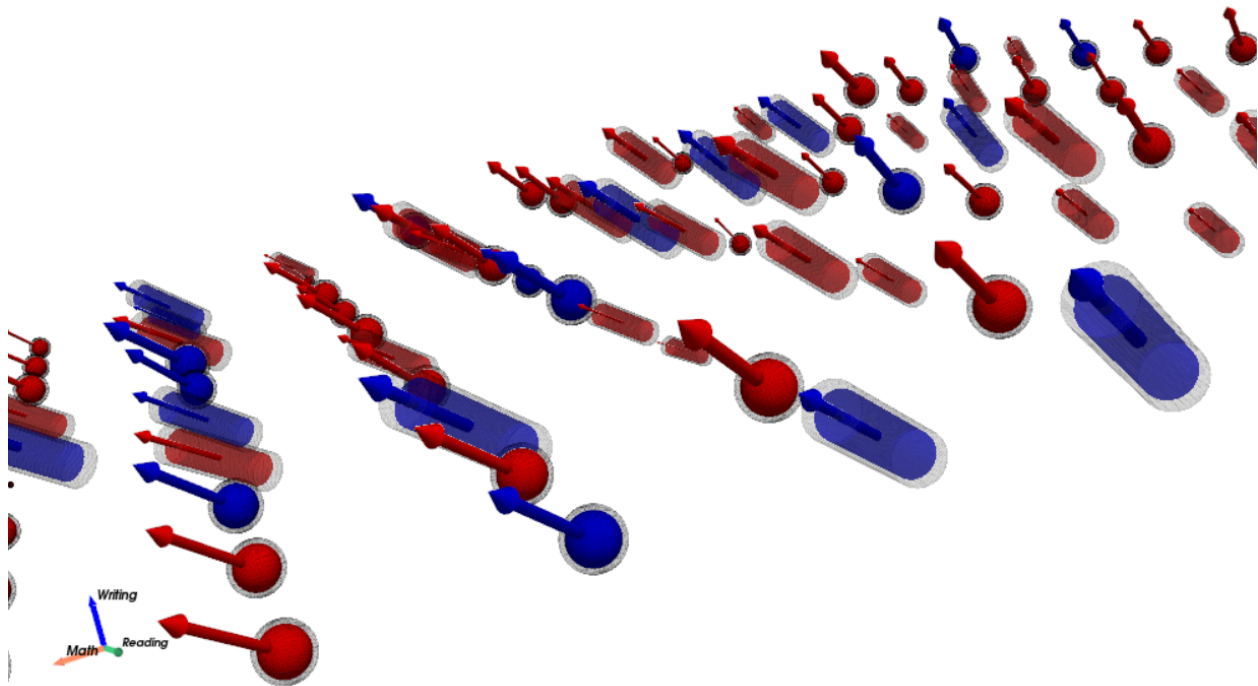


Figure 6: This graph is similar to Figure 4, with a focus on representing uncertainty. The shapes in this figure hold all of the same meanings as in Figure 4. The borders around the shapes in this figure are to represent some of the uncertainties in this representation. Since this data only showcases a subset of students and their performances, the border is meant to show that there are unknowns in this representation.

This Figure helps to show that the data set being represented has some uncertainties. While these figures are meant to be representative of how socio-economic factors can impact students' performance, the full picture isn't represented here. There are many connections that can be drawn from this data, but this data doesn't encompass the set of all students. It only looks at a subset of students.

References:

dataset: <https://www.kaggle.com/datasets/spscientist/students-performance-in-exams/data>